PUSHFORWARD OF AN INJECTIVE MODULE IS ACYCLIC FOR THE GLOBAL SECTION FUNCTORS OF THE TARGET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and ends with the two deductions; no step is referred to the reader.

Editorial note. The counted spans carry 3 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-cohomology/retained-preamble.tex lines 1-81), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-pushforward-injective is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Injective modules have vanishing higher Cech cohomology. Same chapter, Lemma lemma-injective-trivial-cech, invoked at proof lines 1810-1811. Supplies the vanishing of the Cech complex of I on the inverse image covering.
- Vanishing criterion passing from Cech cohomology of coverings to cohomology. Same chapter, Lemma lemma-cech-vanish, invoked at proof lines 1811-1812. Supplies the second statement from the first.
- Right acyclicity of an object for a functor. Derived Categories, Definition derived-definition-derived-functor, invoked at proof lines 1790-1792. Supplies the reformulation in the last sentence of the statement.

Source reference. Original definition 0157: sources/stacks-chow/tags.txt, source lines 1498–1498. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01EV: sources/stacks-cohomology/pinned-original-source.tex, source lines 1631–1631. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01EP: sources/stacks-cohomology/pinned-original-source.tex, source lines 1408–1408. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of ringed spaces. Let $\mathcal{I}$ be an injective $\mathcal{O}_X$-module. Then*

(1) $\check{H}^p(\mathcal{V}, f_*\mathcal{I}) = 0$ *for all $p > 0$ and any open covering $\mathcal{V} : V = \bigcup_{j \in J} V_j$ of Y.*

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(2) $H^p(V, f_*\mathcal{I}) = 0$ *for all* $p > 0$ *and every open* $V \subset Y$.

In other words, $f_*\mathcal{I}$ *is right acyclic for* $\Gamma(V, -)$ *(see Derived Categories, Definition 0157) for any* $V \subset Y$ *open.*

Proof. Set $\mathcal{U} : f^{-1}(V) = \bigcup_{j \in J} f^{-1}(V_j)$. It is an open covering of X and

$$\check{\mathcal{C}}^\bullet(\mathcal{V}, f_*\mathcal{I}) = \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{I}).$$

This is true because

$$f_*\mathcal{I}(V_{j_0 \ldots j_p}) = \mathcal{I}(f^{-1}(V_{j_0 \ldots j_p})) = \mathcal{I}(f^{-1}(V_{j_0}) \cap \ldots \cap f^{-1}(V_{j_p})) = \mathcal{I}(U_{j_0 \ldots j_p}).$$

Thus the first statement of the lemma follows from Lemma 01EP. The second statement follows from the first and Lemma 01EV. □